

POIWG #53 Vehicle Configuration Overview



HUMAN SPACE FLIGHT TECHNICAL INTEGRATION CONTRACT (HSFTIC)

POIWG #53 - ISS External Vehicle Configuration Overview

eCAMMP | MAGIK

Vehicle Configuration Team
Zain (Zainul) Dharmajwala - eCAMMP/MAGIK

October 2023

Vehicle Configuration Overview

eCAMMP & MAGIK Services and Deliverables

- External Configuration Analysis Modeling & Mass Properties (CAMMP) Overview
  - ISS Configuration
  - Clearance Analyses
  - Payload Field of Views (FOVs)
- Manipulator Analysis Graphics and Interactive Kinematics (MAGIK) Overview – Robotics General Information
  - Payloads Analysis
  - EVR Choreography
  - Visualization
- Vehicle Configuration Summary

External CAMMP (eCAMMP) Overview

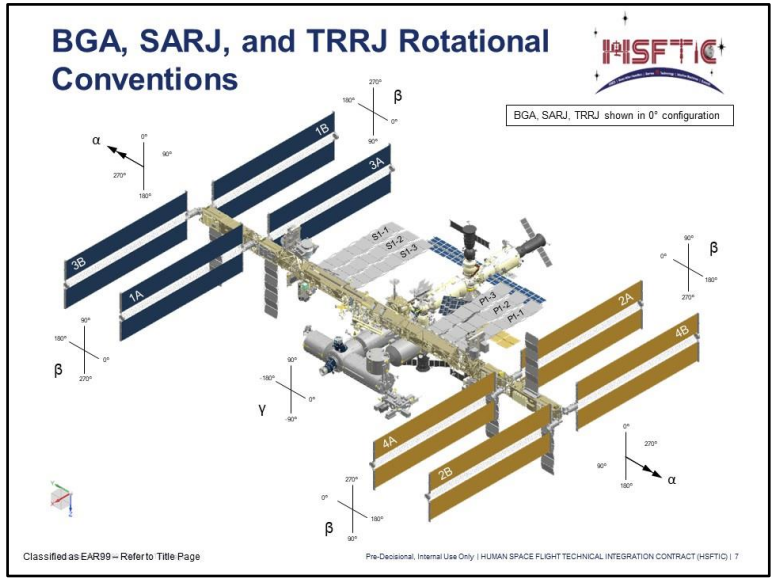
ISS Configuration – eCAMMP General Information

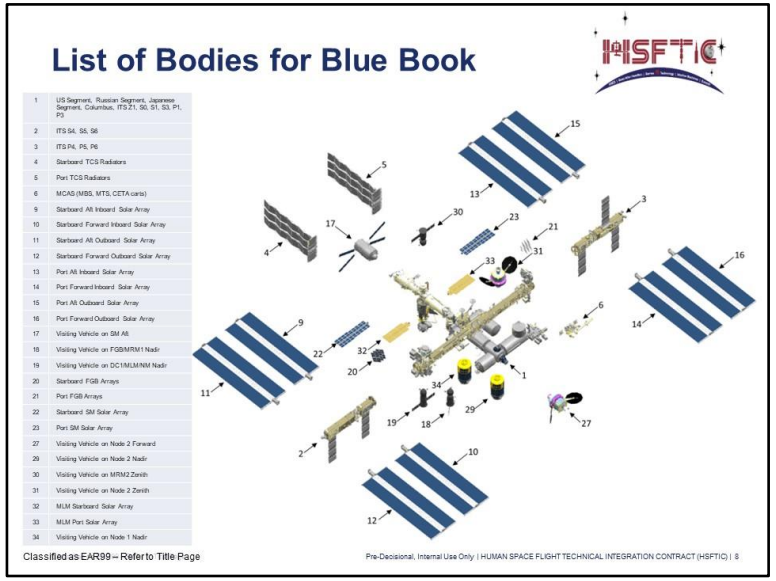
- Tracks, documents, and analyzes past, current, and future external ISS configurations
  - Configuration tracking
    - External Configuration Analysis Tracking Tool (ExCATT) - displays historical external configuration events (Flights, EVAs, EVRs) of ISS at the hardware-item level
    - On-Orbit Configuration Plan (OCP) - displays external integrated configuration events of ISS in a chronological order for past, tactical, and strategic timeframes
  - Clearance analysis including Visiting Vehicle approach, docking, and departure
  - Aerodynamic and Mass Properties (Blue Book)
  - Field-of-View analysis for hardware and payloads
  - Process and generate CAD envelopes for visiting vehicle Jet Firing History data to determine ISS impacts
  - ISS integrated and hardware detail, images and dimensioned drawings
- Data supplied from/to organizations across ISS Program
  - Boeing ISS 3D CAD Model Team (data source for CAD models)
  - OM (OM3/VIPER, OM7/Robotics)
  - FOD (EVA, EVR, ADCO)
  - MER Engineering (GN&C, RPOC, Comm, Robotics)
  - OZ
  - International Partners
  - VV, Payload, Hardware developers
- Action requests are submitted through the eCAMMP Team website

External CAMMP Overview

eCAMMP Analyses - ISS Configuration

- SSP 30219 (Space Station Reference Coordinate Systems)
  - Establishes a set of coordinate systems to be used when reporting data between the International Space Station (ISS) Program Participants
- SSP 50949 (ISS Configuration Document)
  - Details the ISS baseline configuration (external platforms, element configuration, camera port plan, ISS port connectivity, VV support hardware, robotic elements, retroreflector locations, and internal topologies)
- ISS Reference Information
  - SharePoint site providing ISS overview data such as Russian Configuration Numbers, Camera Port Configuration, Current ISS Configuration Images, External Carriers and Payload Site configuration (contained within ISS Reference Packet)
    - External Carrier Site configuration data feeds into IDRD Appendix E
    - Updated on an ongoing basis as planned on-orbit operations are confirmed

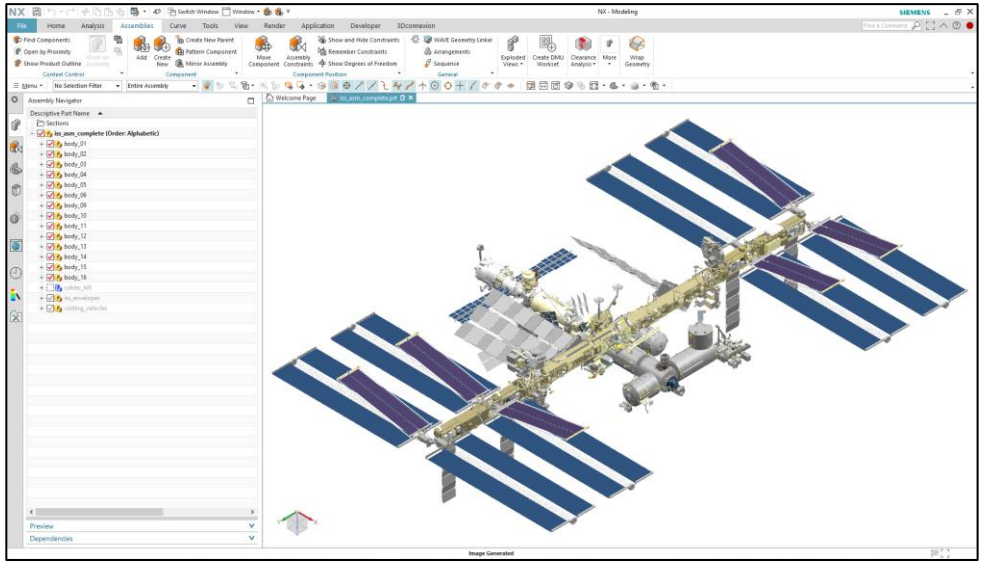


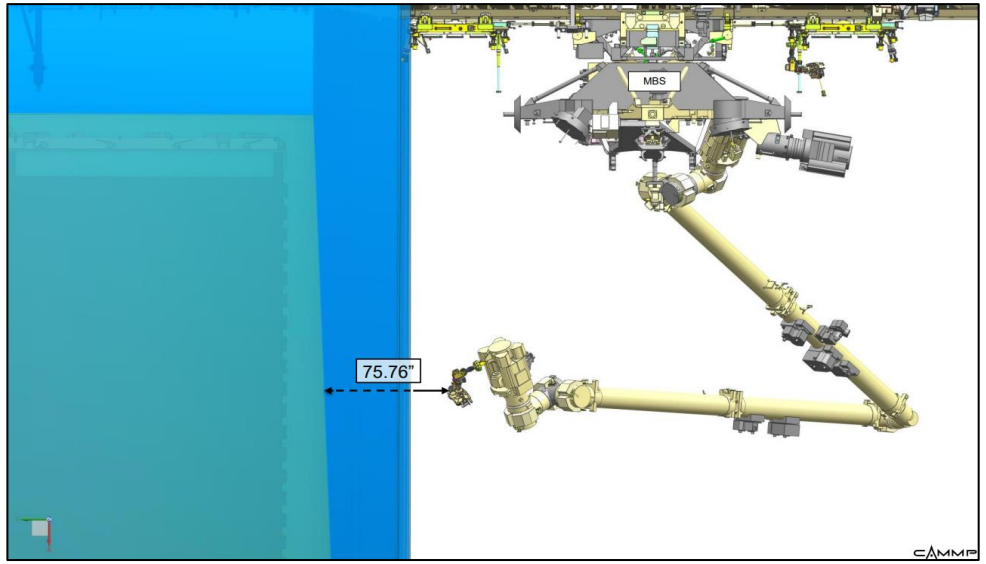


External CAMMP Overview

eCAMMP Analyses – Verification

- 3D CAD Requirements Verification and Feasibility Studies
  - Clearance analysis for new Visiting Vehicles, ISS hardware, and external payloads
  - Evaluate both static and dynamic clearances (via CAMMP generated or client provided envelopes)
    - Robotics Translation Corridor
    - Visiting Vehicle Approach, Docking, and Departure; Berthing Analyses and Viewing
    - Antenna and Camera fields-of-view impingements
    - Feasibility checks for placing systems or payloads on various carrier sites
  - 3D envelope generation through CAD modeling of operational requirements and mathematical point cloud computation of dynamic motion events

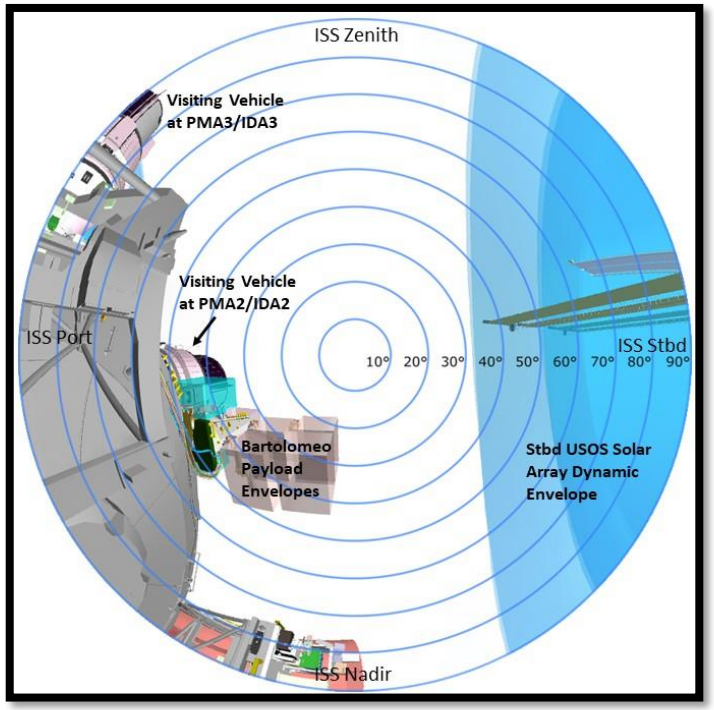




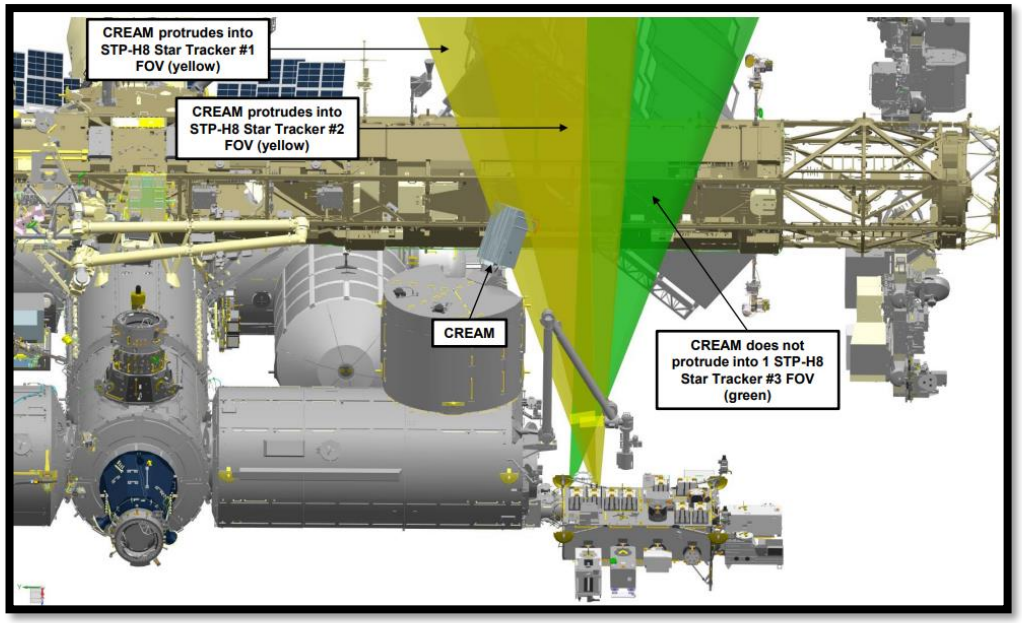
External CAMMP Overview

eCAMMP Analysis – Payload FOVs

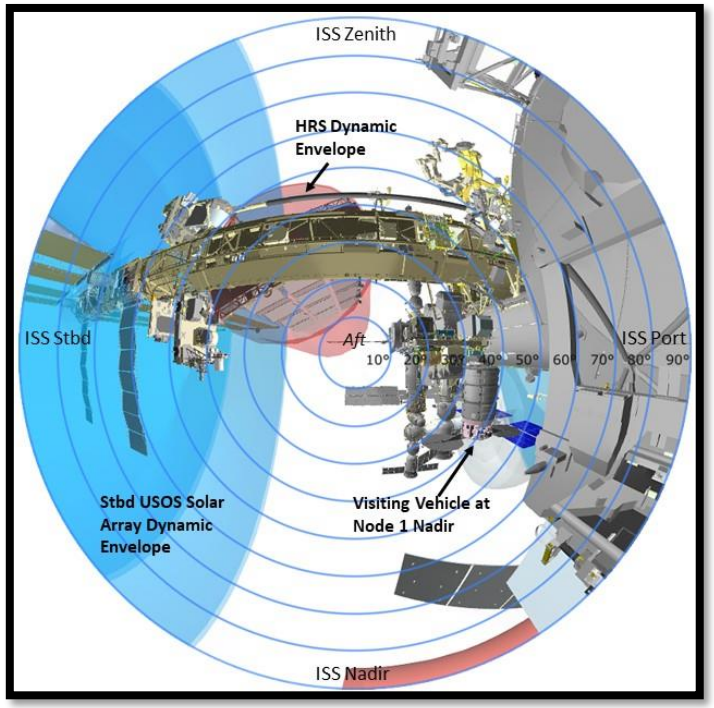
- Generates Field of View (FOV) images for payloads placements on across all ISS external carrier sites for evaluation by the Payload Developers
- FOVs can be provided as NX Camera Fisheye perspectives from sensor locations or modeled as CAD envelopes to identify impingements within view



Fisheye Lens FOV
(ISS Fwd View)



CAD Generated FOV Envelopes
(Multiple Viewing Directions)

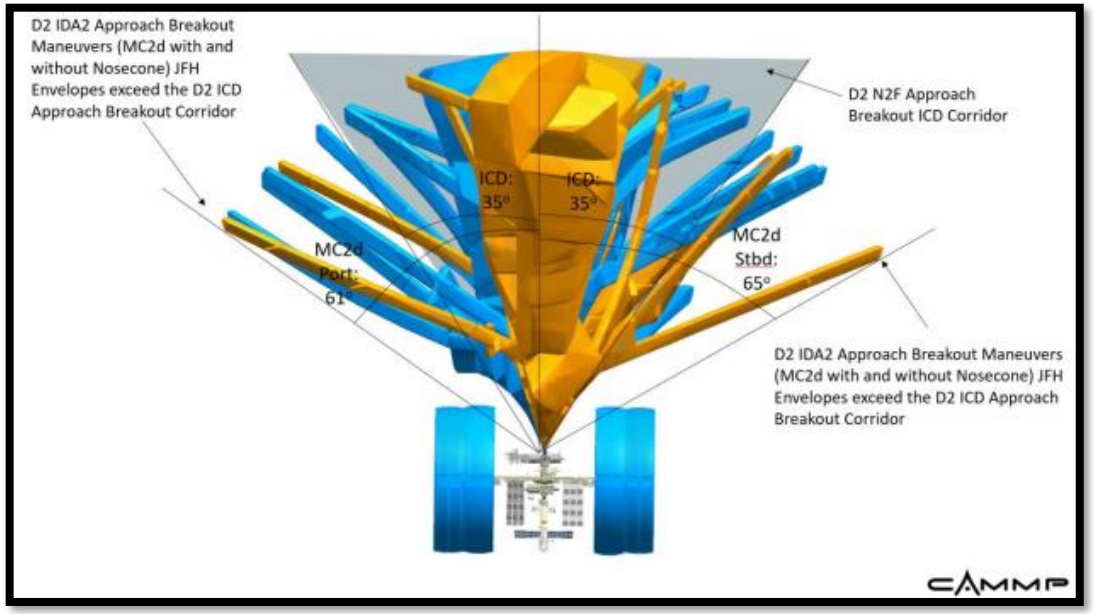


Fisheye Lens FOV
(ISS Fwd View)

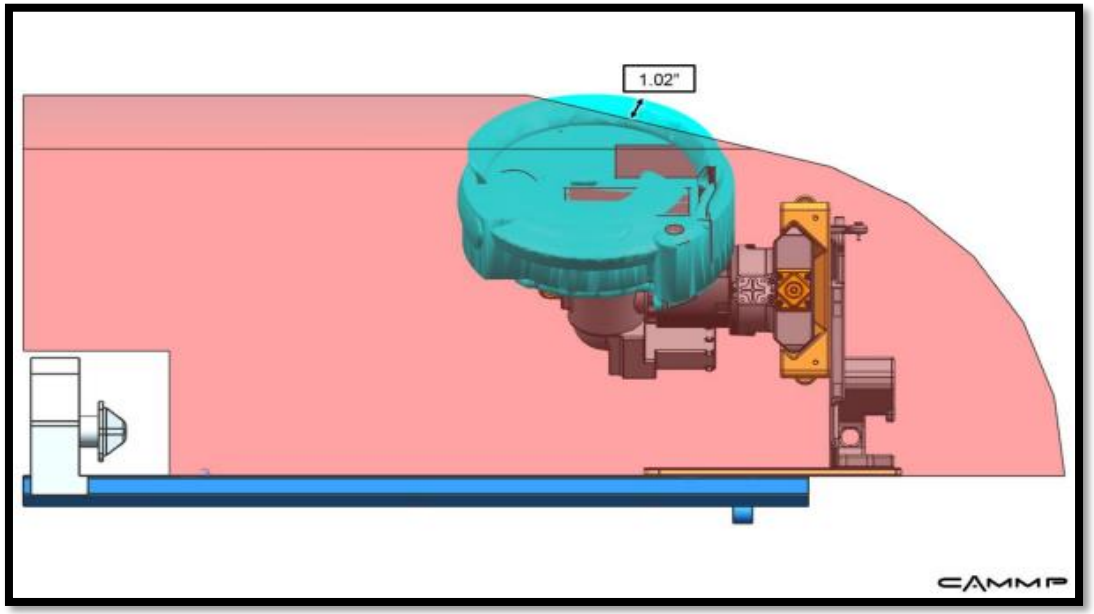
External CAMMP Overview

eCAMMP Analysis – Clearance Analysis

- Provides Clearance Analysis of:
  - VV approach/departure trajectories from nearby ISS Structure
  - Operational motion and Keep-Out-Zones of payloads, hardware, and visiting vehicles modeled as CAD envelopes
  - Verification of ISS internal payload and hardware placement



Jet Firing History
(VV Approach or Departure)

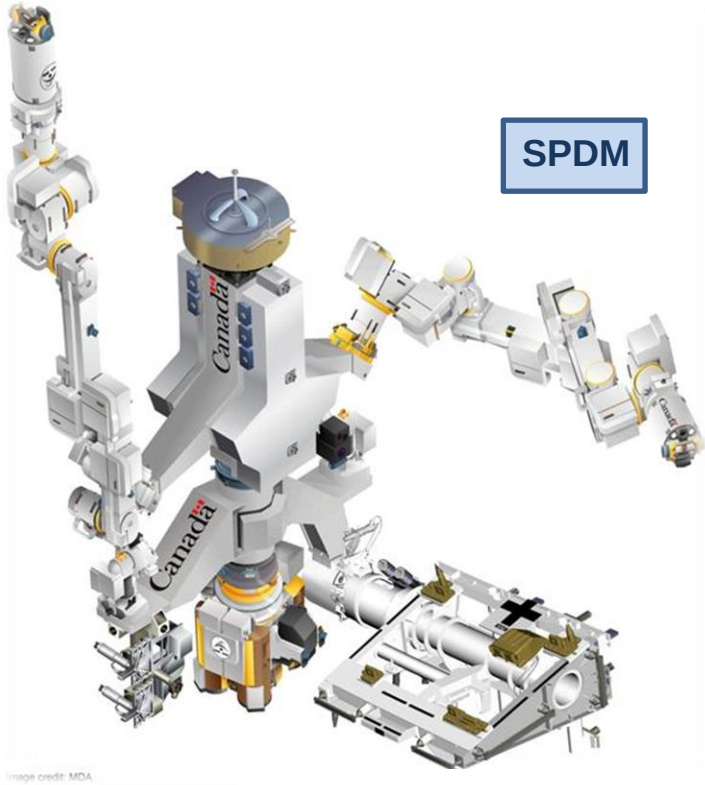


CLPA Tilt Angle Rotational Envelope on
JEM Slide Table

MAGIK Overview

ISS Robotics General Information

- OM7/ISS Robotics Integration leads the coordination with various stakeholders to ensure robotic readiness in support of ISS objectives
- Works closely with FOD through the life cycle to prepare for Payload Operations
- MSS (Mobile Servicing System) and JEM RMS (Japanese Exploration Module Remote Manipulator System) are the primary methods for installation, operation, and disposal of external payloads on the International Space Station
    - JEM RMS, operated by JAXA, is composed of Main Arm and Small Fine Arm (SFA)
    - Used to install and service payloads on the JEM External Facility and JEM Airlock
- Canadian Space Agency (CSA) developed and jointly operated with NASA, MSS is made up of:
    - SSRMS (Space Station Remote Manipulator System)
    - SPDM (Special Purpose Dexterous Manipulator)
    - Mobile Base System (MBS) installed on the Mobile Transporter (MT)

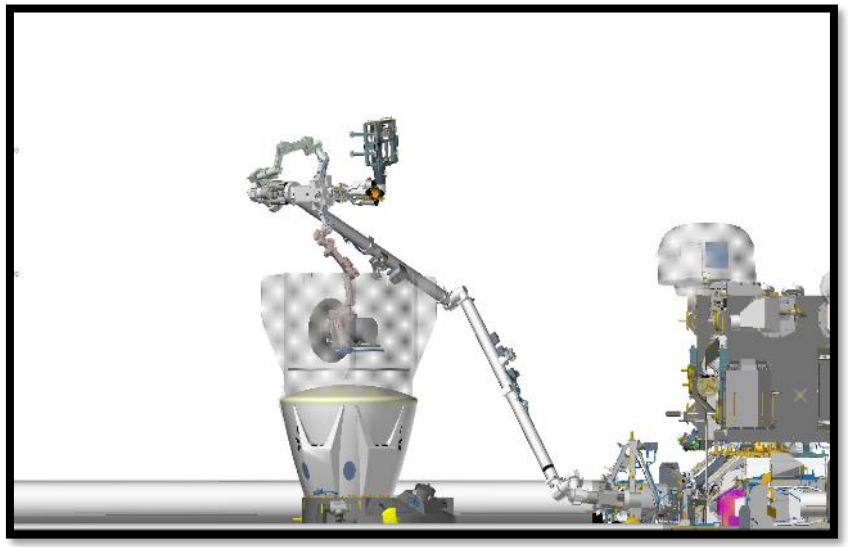




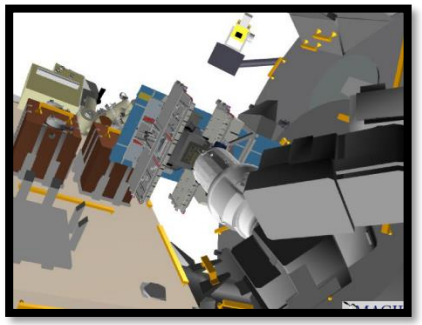
MAGIK Analysis

MAGIK Payloads Analysis

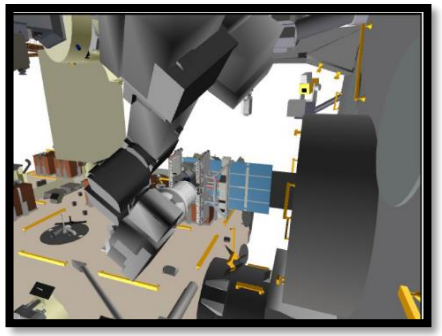
- MAGIK Team supports MSS Integration Panel forums to:
  - Perform kinematic simulation analyses of proposed robotic activities using the SSRMS and SPDM
  - Include trade studies and Action Items (AIs) as required
- Services (assessments) of interest to Payload Developers include:
  - Viewing - ISS & MSS Cameras, Window Viewing
  - Robotic Envelopes - OTCM & SSRMS Approach Envelopes
  - ISS Envelopes - Radiator Sweep Envelopes, Solar Array (SA) Envelopes, Robotic/EVA Translation Corridors
  - SPDM & SSRMS Manipulation - Assess feasibility of payload-specific robotic I/F locations
  - MAGIK assessments can be requested through your payload's Robotic POC or RIM/PIM
- Payloads or Events (i.e. commercial element expansion) involving EVR are run through MAGIK for:
  - Kinematic feasibility
  - Within reach and clearance constraints



RPS Evaluation of Dragon2 Robotics



SPDM CLPA2 Camera View

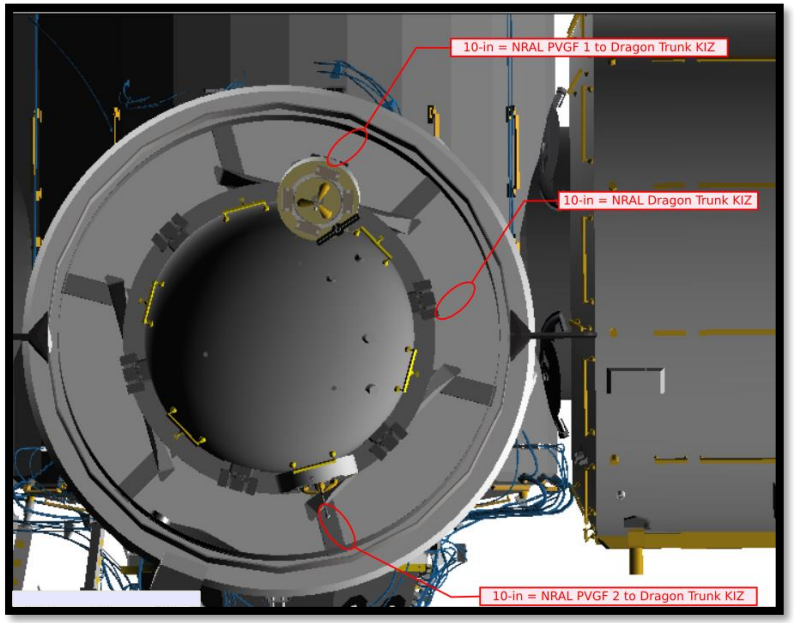


JEM PM Aft Camera View

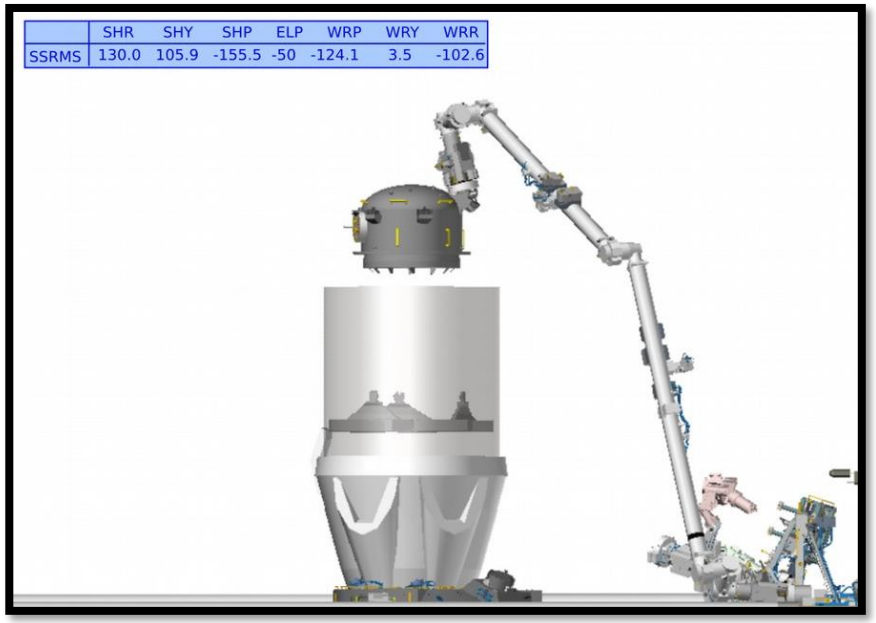
MAGIK EVR Choreography

SpX-21 (L-6) Dragon on N2Z | NR Airlock

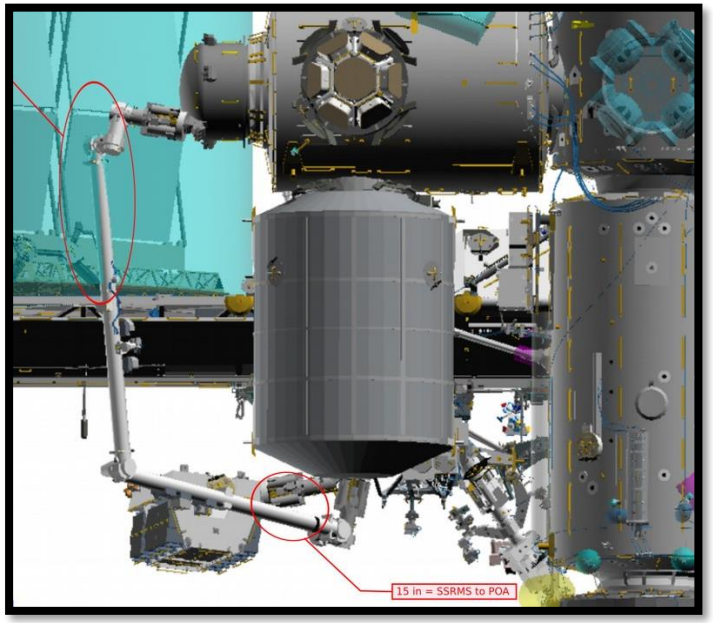
- MAGIK Action Item:
  - NR Airlock will increase utilization capability for science payload exchange and deployment
  - N2Z docked Dragon robotics performed with SSRMS
  - Arm configurations and clearances for end-to-end operations provided to support operations development



NRAL in Dragon Trunk



NRAL Extraction from Dragon Trunk



NRAL installed at N3 Port

Vehicle Configuration Summary

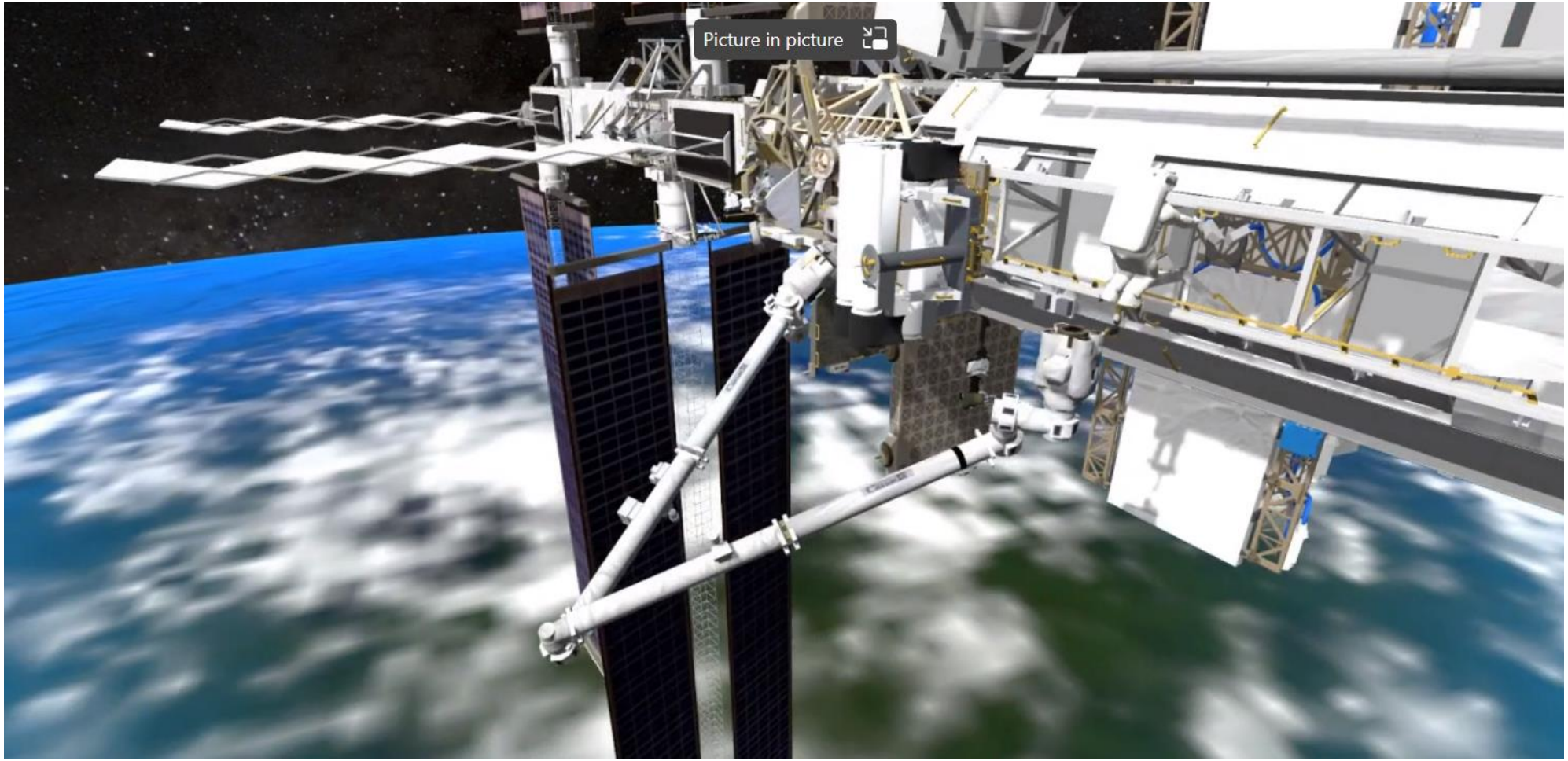
eCAMMP & MAGIK

- eCAMMP
  - External Configuration products include:
    - External Configuration Analysis Tracking Tool (ExCATT)
    - On-Orbit Configuration Plan (OCP)
    - JSC 26557 (On-Orbit Assembly Modeling & Mass Properties Data Book; i.e. Blue Book)
    - SSP 30219 (Space Station Reference Coordinate Systems)
    - SSP 50949 (ISS Configuration Document)
  - Payload analysis for FOV capabilities and placement studies on external carrier sites
  - ISS External Clearance Analyses

- MAGIK
  - Kinematics simulation ensures elements designed to be robotically compatible have appropriate interfaces with MSS hardware and corresponding ISS configuration

MAGIK Visualization

MAGIK Visualization Player (MVP) Video Sequence Creation



MAGIK IROSA Installation Movie

https://ntrs.nasa.gov/citations/20220018728



POIWG #53 - ISS Internal Vehicle Configuration Overview

iCAMMP/IPiP

Vehicle Configuration Team
Skyler VonDeben - iCAMMP
Cameryn Lopez - IPiP

October 2023

Vehicle Configuration Overview

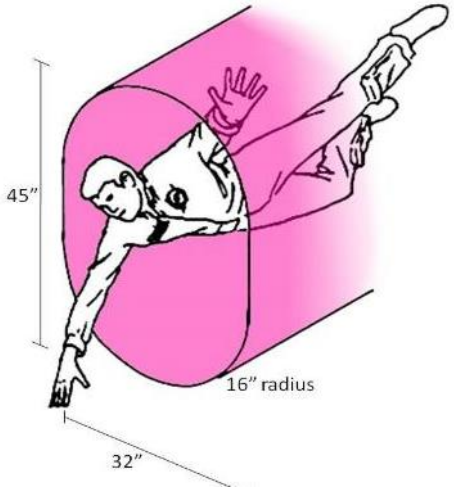
iCAMMP Services and Deliverables

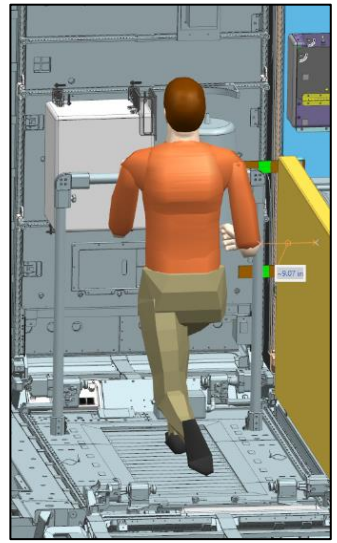
- Internal Configuration Analysis Modeling & Mass Properties (CAMMP) Overview
  - Analyses – Types & Processes
  - Topology Hardware & GGR&C Analysis
  - ISS Plug-in Plan (IPiP) Products, Procedures, and Timeline
- Vehicle Configuration Summary

Internal CAMMP Overview

Services & Deliverables

- iCAMMP/IVCWG Definitions
  - Maintains 3D CAD models of the interior configuration of ISS
  - Performs analyses on the planned and current configuration
  - Internal Volume & Configuration Working Group (IVCWG) is the ISS community working group that meets monthly to discuss internal configuration changes and requirements
- Types of analyses performed by iCAMMP include:
  - I-4.5 mo. and I-1.5 mo. Increment Configuration Analysis
  - Placement Analysis for systems hardware
  - Cargo Translation Analysis
  - Stowage Capability Analysis
- Additional iCAMMP Responsibilities
  - Document and maintain SSP 50261-01 GGR&C Section 13
  - Document and maintain USOS Rack Topology (baselined in IDRD)
  - Integration activities for internal volume configuration

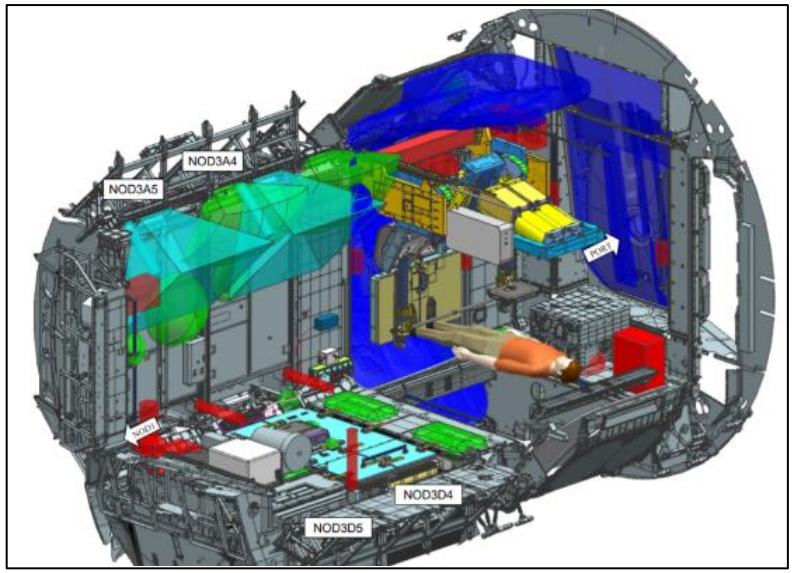




Internal CAMMP Overview

iCAMMP Hardware Analyses

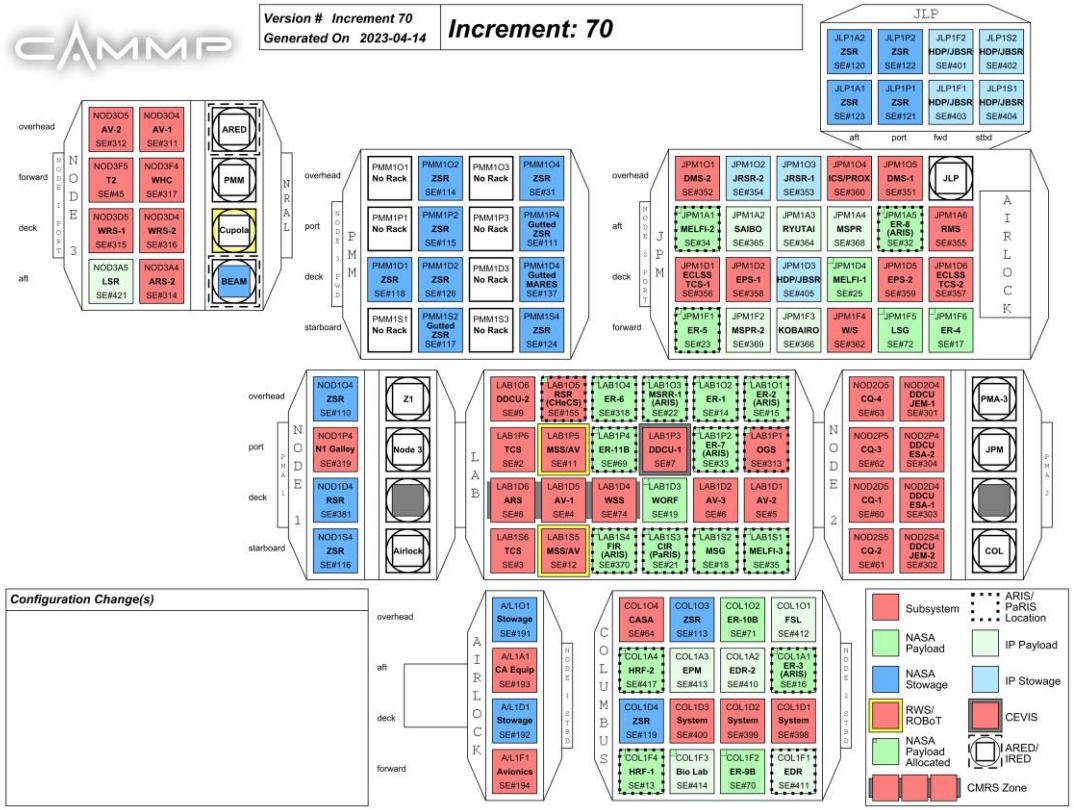
- Placement Analyses
  - Performed as requested to identify locations for systems hardware installation or relocation
  - Requestor provides requirements/constraints and multiple locations are assessed for GGR&C Section 13 requirements
- Increment Analyses
  - Performed at Increment minus 4.5 months (I-4.5m) and at I-1.5m
  - Verifies compliance of planned internal configuration of the US Lab, US Airlock, PMAs, and Nodes 1, 2, and 3 to GGR&C Section 13
  - Multiple organizations' data and previously completed analyses are used to build the Increment configuration models
- Stowage Analyses
  - Performed to analyze stowage configurations supplementing hardware placement or protrusion assessments
  - Assessments can include:
    - Analyzing volumetric capability
    - Packing efficiency studies
    - Images for stowage visualization



Internal CAMMP Overview

iCAMMP Configuration and Requirements

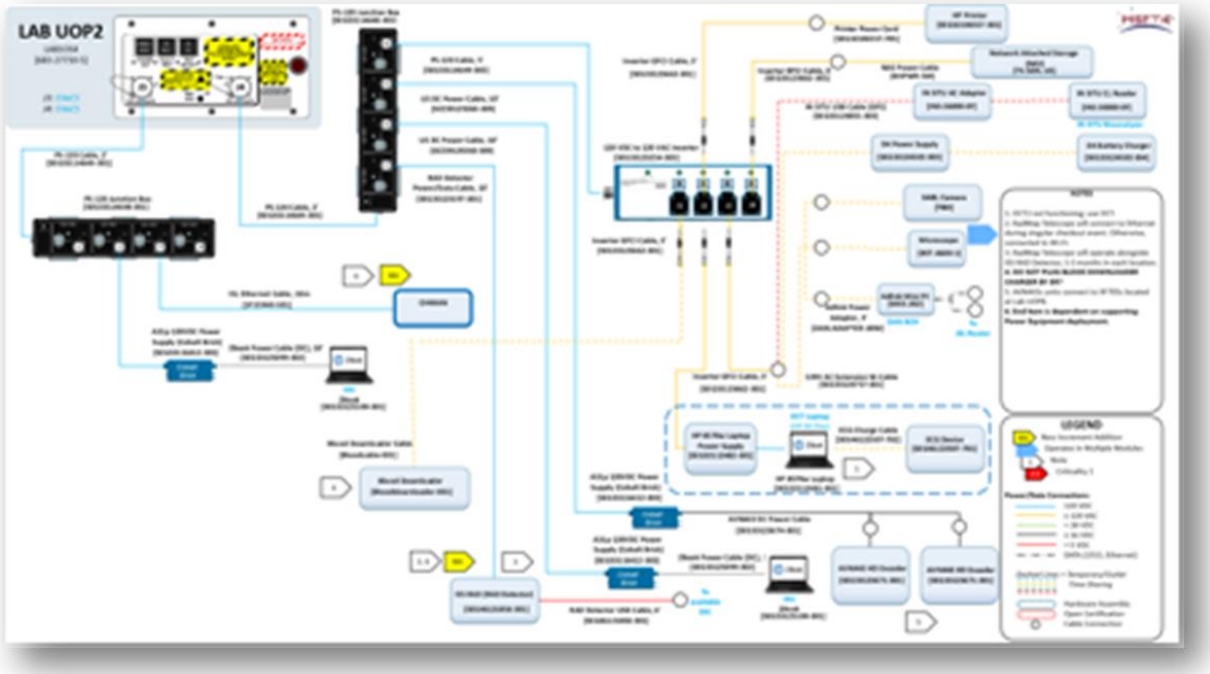
- USOS Rack Topologies
  - Maintain Internal Rack Topologies for all planned configurations
  - Changes are documented in the Increment Definition and Requirements Document (IDRD)
- Generic Groundrules, Requirements, and Constraints (GGR&C)
  - iCAMMP co-owns GGR&C Section 13 Interior Volume Configuration
  - GGR&C ensures a safe, operable, and habitable environment for crew to live on ISS
  - iCAMMP leads ISS Video Survey coordination every 6 months to identify and mitigate on-orbit unplanned violations



ISS Plug-in Plan (IPiP) Overview

IPiP Team Products and Purpose

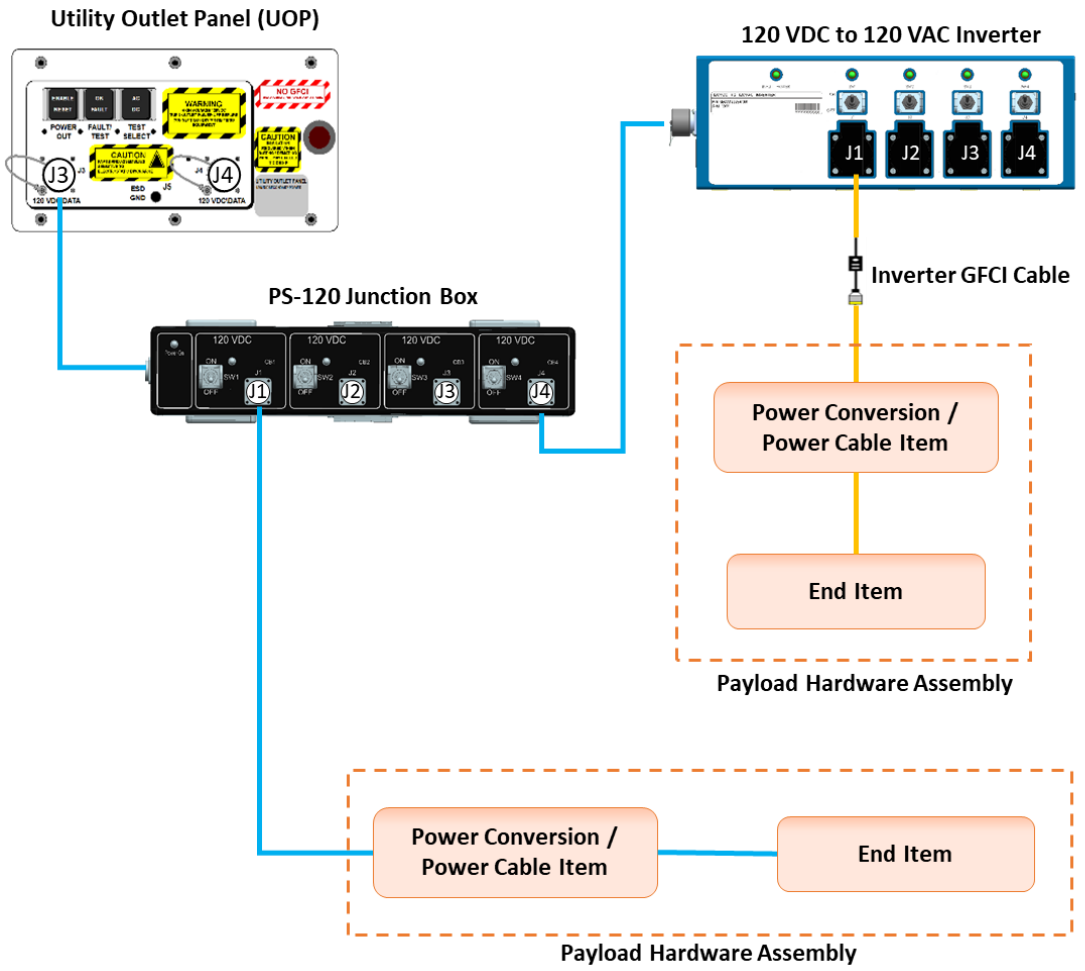
- Coordinates and develops the integrated plug-in plan for Payload and Systems hardware power requirements
- Power requirements are collected through IPiP forms
  - IPiP Change Evaluation Forms (CEFs) in OZ Requirements Baseline Integration Tool (ORBIT) Research Investigation and Facility Database (RIFD) for Payloads
  - IPiP Requirement Request Forms (RRFs) for Systems
- Increment Baseline Schematic is delivered to FOD Plug-in Port Utilization Task Officer (PLUTO) for real-time execution



ISS Plug-in Plan (IPiP) Overview

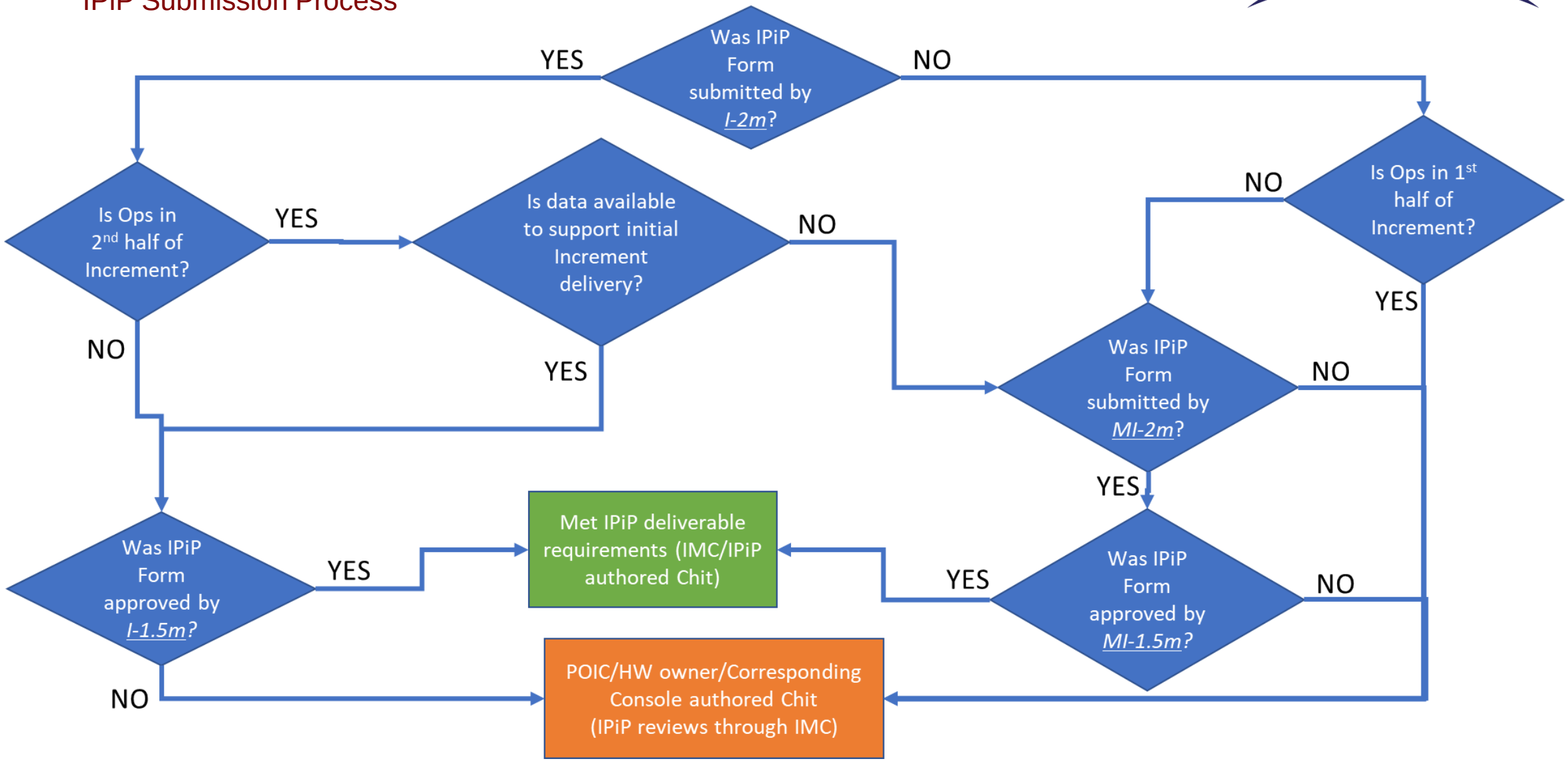
IPiP Hardware

- Plug-in Plan (PiP) hardware draws power from Utility Outlet Panels (UOPs) and Standard Utility Panels (SUPs)
- Common Station Support Equipment (SSE) requested via PIA and RP CEFs include:
  - PS-120 Junction Box
  - 120VDC/VAC Inverter
  - KU-Band Power Supply
  - Multiport USB Chargers
- Payload Developer (PD)-provided hardware assembly may include power conversion item, cable item, or both



ISS Plug-in Plan (IPiP) Overview

IPiP Submission Process



ISS Plug-in Plan (IPiP) Overview

IPiP Timeline

Mid-Increment 70 Schedule

First Call for IPiP Forms [IMC Coordination] (MI70 - 3m)	IPiP Forms Final Deadline (MI70 – 1.5m)	IPiP DRAFT Baseline Delivery for PLUTO & IP Review (MI70 - 1.5m)	IPiP, PLUTO, & IP Review of Baseline [OP/CO] (MI70 – 3w)	Post IPiP Baseline on IPiP Website and Deliver CoFR Letter (MI70 - 2w)
9/28/23	11/17/23	12/1/2023	12/8/2023	12/15/2023

Initial Increment 71 Schedule

First Call for IPiP Forms [IMT Kickoff] (I70 - 8m)	Increment Operations Review [IOR] (I70 – 4m)	IPiP Forms Final Deadline (I70 - 2m)	IPiP DRAFT Baseline Delivery for PLUTO & IP Review (I70 - 1.5m)	IPiP, PLUTO, & IP Review of Baseline [OP/CO] (I70 - 1m)	Post IPiP Baseline on IPiP Website and Deliver CoFR Letter (I70 - 4w)
9/19/23	TBD	1/25/24	2/8/24	2/22/24	2/26/24

Vehicle Configuration Summary

iCAMMP & IPiP

- iCAMMP
  - Performs verification of GGR&C Section 13 for integrated USOS internal configuration
  - Placement analyses of systems hardware
  - Internal configuration products include Increment Analysis, USOS Rack Topology, Stowage Analysis
- IPiP develops the increment Plug-in Plan for UOP/SUPs throughout ISS by Increment
  - Delivers Baseline Schematic to PLUTO for real-time implementation
  - RIFD CEFs are necessary for Payload power allocation requests
  - Authors PIP Tool Dataset Chits if requirements are met
  - Walkthrough and other resources are available on the OM Website